HW6

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2026-03-09

Problem 1: Concept Review

(a)

Best subset selection will produce a collection of p models M1, M2, . . . , Mp. These represent the 'best' model of each size (where 'best' here is defined as the model with the smallest RSS).

(i) True or False

Is it true that the model identified as Mk+1 must contain a subset of the predictors found in Mk? In other words, is it true that if M1 : Y ~ X1, then M2 must also contain X1. And if M2 contains X1 and X2, then M3 must also contain X1 and X2?

False. Since best subset selection evaluates all possible combinations of predictors and picks the best one, there is no guarantee that it will be a subset.

(b)

Same question as part (a) but instead of best subset selection, we now carry out forward selection.

True. Forward selection works by adding the best predictor for each size. So Mk+1 is Mk + one more predictor.

(c) (practice)

Suppose we perform best subset selection, forward selection, and backward selection on a training data set.

Best subset selection produces a model with k predictors for each k = 1, . . . , p. Denote this model by Mk,subset. For example, when k = 5, M5,subset is the best 5-predictor model according to best subset selection.

Similarly, forward selection produces a model with k predictors, denoted by Mk,forward, and backward selection produces a model with k predictors, denoted by Mk,backward.

For each case below, state which of the three models would be expected to have the smallest training MSE, and briefly explain why.

(i) (practice) k = 1

Solution: For k = 1, subset has the same model as forward stepwise and therefore the same RSS (and training MSE). Backward stepwise may have a different model, and a worse RSS.

(ii) (practice)k ϵ {2, . . . , p - 2}

Best subset selection. Since it's guaranteed to find the lowest RSS for any k it will be the best. Forward and backward only make the best choice for each size (greedy).

(iii) (practice)k = p - 1

Best subset selection. The same logic as part ii.

(iv) (practice)k = p

All three are equal. There's only one model that contains all p predictors so they will all be the same.

(d) (practice)

Same setup as part (c). For a given k, which of these three models has the smallest test MSE? Explain your answer.

(i) k = 1

Solution: Subset and forward stepwise have the same model when k = 1, so they have the same test MSE. Backward selection may produce a different model, so its test MSE may differ.

(ii) k = p

All three are equal. There is only one model with all p predictors so they will all be the same.

(iii) k = p - 1

Cannot say. Best subset selection will have the best training MSE but that doesn't mean it will have the lowest test MSE.

Problem 2: Simulation Studies

We have seen that as the number of features used in a model increases, the training error will necessarily decrease, but the test error may not. We will now explore this in a simulated data set. Start by generating a data set with p = 20 features, n = 1000 observations, and an associated response vector Y generated according to the model:

Y = β_0 + β_1 X1 + β_2 X2 + . . . β_p Xp + ϵ

You can use this code to get started:

```
p = 20
n = 1000
error = rnorm(n,0,1)
beta = c(2, rep(0,5),1.2,3,-2,1.5,7,rep(0,5),3,4,1,9,2)
Xmat = matrix(rnorm(n*20),n,20)
X = cbind(rep(1,n),Xmat)
Y = X%*%beta + error
data = data.frame(Y,Xmat)
```

Note that X%%*beta is just doing matrix multiplication: beta0 + beta1*X1 + beta2*X2 + ...

beta20*X20.

(a)

Split your data set into a training set containing 100 observations and a test set containing 900 observations.

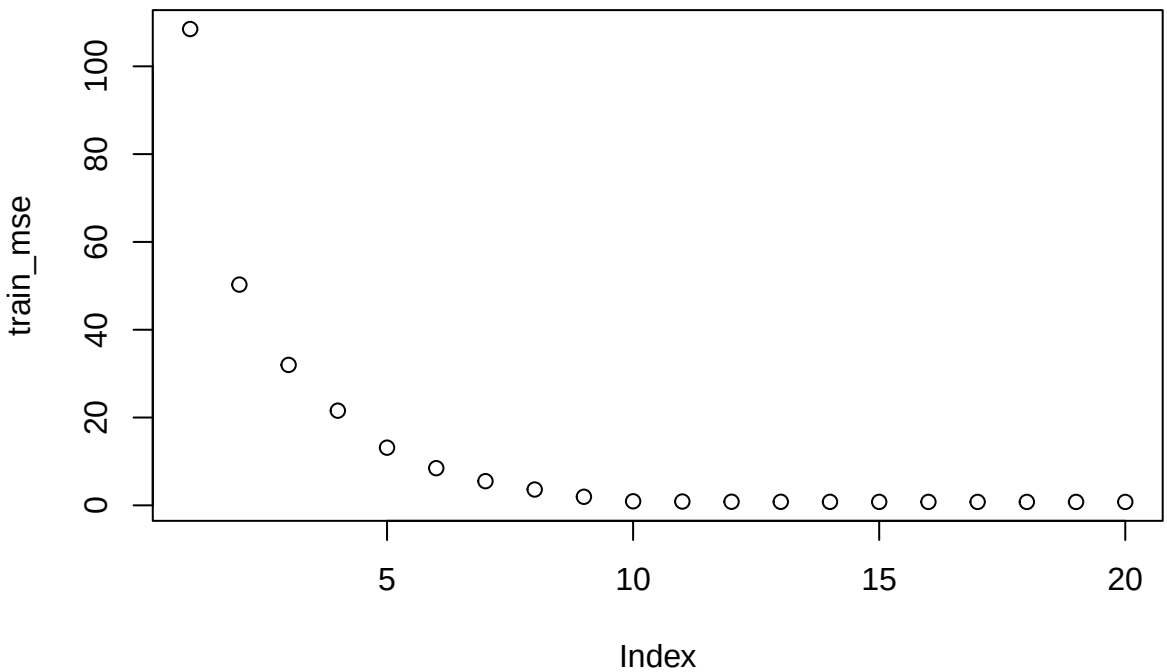
```
set.seed(123)

train_indices = sample(1:n, 100)
```

```r
train_data = data[train_indices, ]
test_data = data[-train_indices, ]
```

(b)

Perform best subset selection on the training set only, and plot the training set MSE associated with the best model of each size. Hint: The training MSE can be computed from the model output. If the fitted model is stored as fit, you can extract it using mean(fit$rss).

```r
library(leaps)

regfit_full = regsubsets(Y ~ ., data = train_data, nvmax = 20)
reg_summary = summary(regfit_full)

train_mse = reg_summary$rss / 100

plot(train_mse)
```

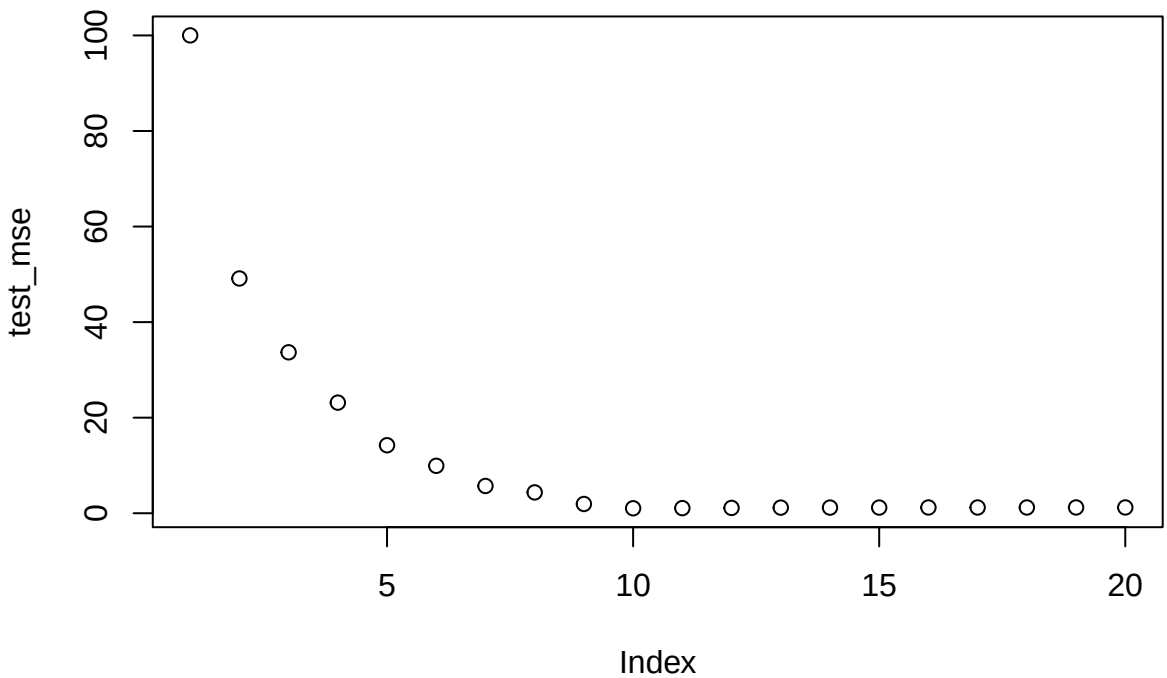


(c)

Plot the test MSE associated with the best model of each size.

```r
test_mse = rep(NA, 20)
test_mat = model.matrix(Y ~ ., data = test_data)

for (i in 1:20) {

  coefi = coef(regfit_full, id = i)
  pred = test_mat[, names(coefi)] %*% coefi
  test_mse[i] = mean((test_data$Y - pred)^2)

}
```

```r
plot(test_mse)
```



(d)

For which model size does the test MSE take on its minimum values? Comment on your results.

```r
print(which.min(test_mse))
```

```
## [1] 10
```

The min test MSE comes from 10. This makes sense because the true model only has 10 non-zero predictors so anything more than 10 begins to overfit the data.

(e)

How does the model at which the test MSE is minimized compare to the true model used to generate the data? Comment on how well the estimated regression coefficients match the true values.

Note: The true model contains only 10 predictors with nonzero coefficients. The remaining 10 predictors have true coefficient value equal to zero.

```r
# Y is the true model
diff <- coef(regfit_full, id = 10) - beta[beta != 0]

cbind(coef(regfit_full, id = 10), beta[beta != 0], diff)
```

```
##                                   diff
## (Intercept)  2.164150  2.0  0.16414996
## X6           1.282582  1.2  0.08258164
## X7           3.022102  3.0  0.02210211
## X8          -2.059768 -2.0 -0.05976814
## X9           1.425878  1.5 -0.07412162
## X10          6.907710  7.0 -0.09228964
## X16          3.058654  3.0  0.05865419
## X17          4.048671  4.0  0.04867073
```

```
## X18          1.061106  1.0  0.06110559
## X19          8.967143  9.0 -0.03285678
## X20          1.955628  2.0 -0.04437191
```

I have visualized the differences with the diff column of the chart. As seen in the chart. The biggest difference is ~0.2.

Problem 3: Forward and backward selection (practice)

We will use the College data set in the ISLR2 library to predict the number of applications (Apps) each university received. Use all remaining variables as predictors.

(a) (practice)

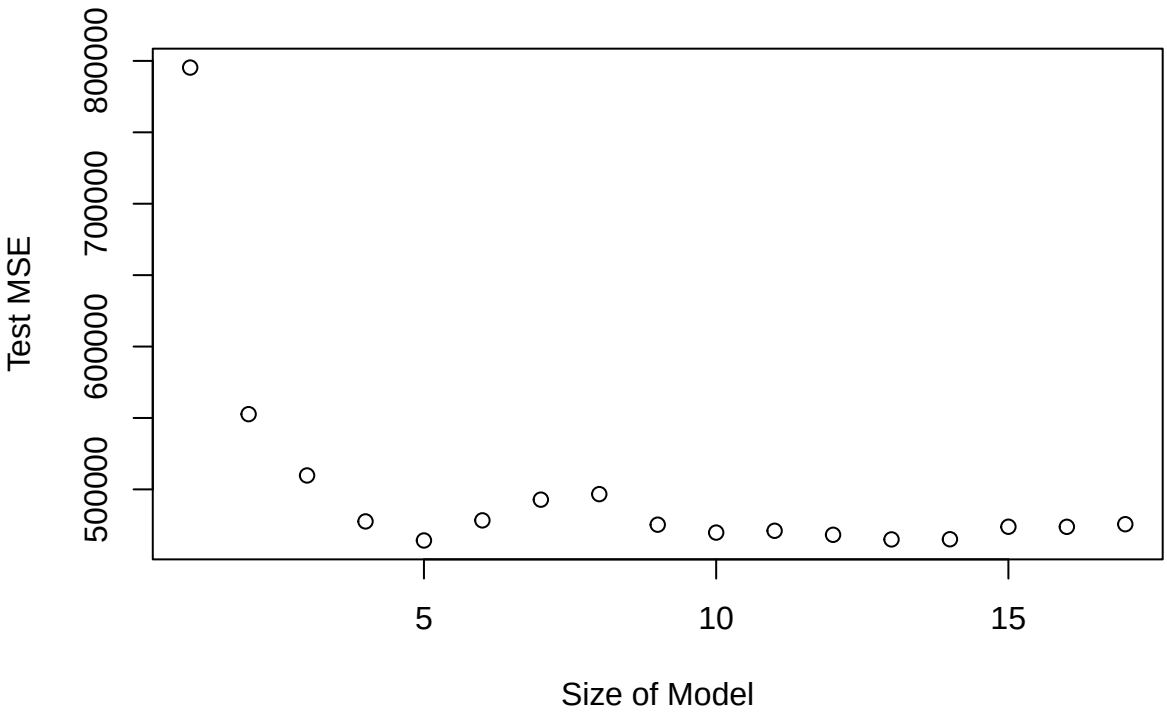
Randomly split the data set so that 90% of the data belong to the training set and the remaining 10% belong to the test set. Implement forward and backward selection on the training set only. Report the test MSE for all models (M1, M2, . . . , Mp) based on forward selection. Which model minimizes test MSE?

Report the test MSE for all models (M1, M2, . . . , Mp) based on backward selection. Which model minimizes test MSE?

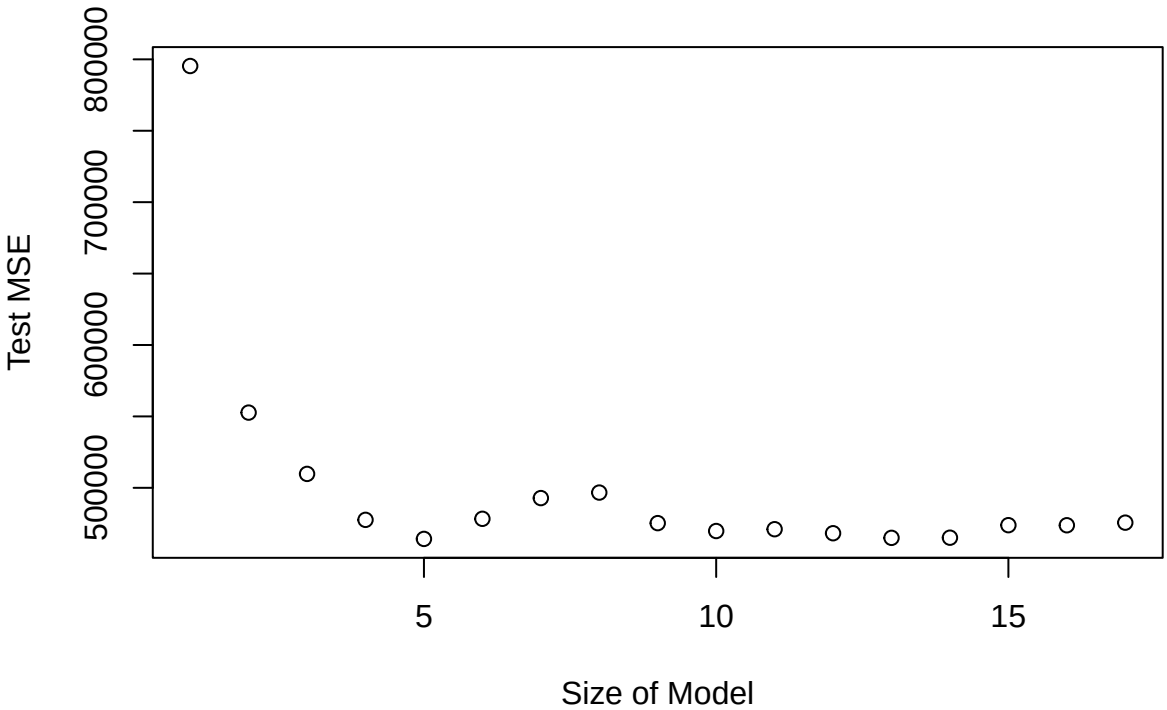
Do both algorithms lead you to the same model based on test MSE?

```r
library(ISLR2)
library(leaps)

set.seed(123)

n <- nrow(College)
p <- ncol(College) - 1
train_id = sample(1:n, 0.9 * n)
train = College[train_id, ]
test = College[-train_id, ]


regfit_fwd <- regsubsets(Apps ~ ., data = train, nvmax = p, nbest = 1, method = "forward")
regfit_bwd <- regsubsets(Apps ~ ., data = train, nvmax = p, nbest = 1, method = "backward")


regfit_fwd_sum <- summary(regfit_fwd)
regfit_bwd_sum <- summary(regfit_bwd)

forward_test_matrix <- model.matrix(Apps ~ ., data = test)
backward_test_matrix <- model.matrix(Apps ~ ., data = test)
forward_test_mse <- rep(NA, p)
backward_test_mse <- rep(NA, p)

for (i in 1:p) {

  coefi = coef(regfit_fwd, id = i)
  pred = forward_test_matrix[, names(coefi)] %*% coefi
  forward_test_mse[i] = mean((test$Apps - pred)^2)


  # Backward Calculation
```

```r
  coef_bwd = coef(regfit_bwd, id = i)
  pred_bwd = backward_test_matrix[, names(coef_bwd)] %*% coef_bwd
  backward_test_mse[i] = mean((test$Apps - pred_bwd)^2)
}

num_predictors <- rowSums(regfit_fwd_sum$which) - 1


plot(num_predictors, forward_test_mse, xlab = "Size of Model", ylab = "Test MSE")
```




```r
plot(num_predictors, backward_test_mse, xlab = "Size of Model", ylab = "Test MSE")
```




```r
cat("Best Forward Model size:", which.min(forward_test_mse), "\n")
```

```
## Best Forward Model size: 5
```

```r
cat("Best Backward Model size:", which.min(backward_test_mse), "\n")
```

```
## Best Backward Model size: 5
```

Based on my results, 5 has the best Test MSE for both forward and backward selection.

(b) (practice)

Run forward and backward selection on the full dataset. For each approach, report the best model based on AIC. Do they lead you to the same model based on AIC?

```r
fit_fwd_full <- regsubsets(Apps ~ ., data = College, nvmax = p, method = "forward")
fit_bwd_full <- regsubsets(Apps ~ ., data = College, nvmax = p, method = "backward")

sum_fwd_full <- summary(fit_fwd_full)
sum_bwd_full <- summary(fit_bwd_full)

rss_fwd <- sum_fwd_full$rss

AIC_fwd <- n * log(rss_fwd / n) + 2 * p

rss_bwd <- sum_bwd_full$rss

AIC_bwd <- n * log(rss_bwd / n) + 2 * p

AIC_fwd
```

```
##  [1] 11154.94 10948.10 10924.20 10895.82 10870.79 10860.67 10850.09 10839.67
##  [9] 10829.80 10822.98 10817.64 10815.53 10814.13 10813.68 10813.41 10813.40
## [17] 10813.40
```

```r
AIC_bwd
```

```
##  [1] 11154.94 10948.10 10924.20 10895.82 10870.79 10860.67 10851.49 10838.41
##  [9] 10829.80 10822.98 10817.64 10815.53 10814.13 10813.68 10813.41 10813.40
## [17] 10813.40
```

```r
best_fwd_aic <- which.min(sum_fwd_full$cp)
best_bwd_aic <- which.min(sum_bwd_full$cp)

cat("Best Forward model size (AIC):", best_fwd_aic, "\n")
```

```
## Best Forward model size (AIC): 12
```

```r
cat("Best Backward model size (AIC):", best_bwd_aic, "\n")
```

```
## Best Backward model size (AIC): 12
```

```r
vars_fwd <- names(coef(fit_fwd_full, best_fwd_aic))
vars_bwd <- names(coef(fit_bwd_full, best_bwd_aic))

vars_fwd
```

```
##  [1] "(Intercept)" "PrivateYes"  "Accept"      "Enroll"      "Top10perc"
##  [6] "Top25perc"   "F.Undergrad" "P.Undergrad" "Outstate"    "Room.Board"
## [11] "PhD"         "Expend"      "Grad.Rate"
```

```r
vars_bwd
```

```
##  [1] "(Intercept)" "PrivateYes"  "Accept"      "Enroll"      "Top10perc"
##  [6] "Top25perc"   "F.Undergrad" "P.Undergrad" "Outstate"    "Room.Board"
## [11] "PhD"         "Expend"      "Grad.Rate"
```

The models are identical.

(c) (practice)

Which is the final model that you would pick? If both (a) and (b) lead you to the same model, then this is easy. Otherwise, there is no absolute right answer. Try to justify your choice using some ideas/concepts you've learned from this class.

Since both models are identical, either would work.

Problem 4: A Puzzling Problem

When fitting a linear regression model on a data set, you encounter the following R output. You notice there is something strange about the results. Point out what is strange in this output and explain clearly how this could happen.

Call: lm(formula = y ~ x1 + x2)

The F statistic p-value is saying that at least one of the predictors is significant but the individual p values are telling us that neither predictor is significant.

The reason for this is high multicollinearity meaning x1 and x2 are highly correlated with each other. When models are highly correlated, the model struggles to see the individual importance of each predictor.

Problem 5: Interaction Terms

We will use the Credit dataset for this problem. It is part of the library ISLR2

(a)

This data set contains a few categorical predictors. As we already discussed in lecture, these predictors should be stored as factors so that R can handle them properly. Using the str function, check that all the qualitative predictors in our dataset are stored correctly in R as factors. Copy and paste your output.

```r
library(ISLR2)

str(Credit)
```

```
## 'data.frame':    400 obs. of  11 variables:
##  $ Income   : num  14.9 106 104.6 148.9 55.9 ...
##  $ Limit    : num  3606 6645 7075 9504 4897 ...
##  $ Rating   : num  283 483 514 681 357 569 259 512 266 491 ...
##  $ Cards    : num  2 3 4 3 2 4 2 2 5 3 ...
##  $ Age      : num  34 82 71 36 68 77 37 87 66 41 ...
##  $ Education: num  11 15 11 11 16 10 12 9 13 19 ...
##  $ Own      : Factor w/ 2 levels "No","Yes": 1 2 1 2 1 1 2 1 2 2 ...
##  $ Student  : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 1 1 1 2 ...
##  $ Married  : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 1 1 1 1 2 ...
##  $ Region   : Factor w/ 3 levels "East","South",..: 2 3 3 3 2 2 1 3 2 1 ...
##  $ Balance  : num  333 903 580 964 331 ...
```

Own, Student, Married, and Region are factors.

(b)

Fit a model with the response (Y) as credit card balance and X1= Income and X2 = Student as the predictors. Call this model fit. Summarize your output.

```r
?Credit
fit <- lm(Balance ~ Income + Student, data = Credit)


summary(fit)
```

```
##
## Call:
## lm(formula = Balance ~ Income + Student, data = Credit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -762.37 -331.38  -45.04  323.60  818.28
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 211.1430    32.4572   6.505 2.34e-10 ***
## Income        5.9843     0.5566  10.751  < 2e-16 ***
## StudentYes  382.6705    65.3108   5.859 9.78e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 391.8 on 397 degrees of freedom
## Multiple R-squared:  0.2775, Adjusted R-squared:  0.2738
## F-statistic: 76.22 on 2 and 397 DF,  p-value: < 2.2e-16
```

(c)

Based on our results from part (b), write out the fitted model for

i students

students = 211.1430 + 5.9843*income + 382.6705*1

ii non-students.

non_students = 211.1430 + 5.9843*income + 382.6705*0

(d)

Interpret the regression coefficient related to Income for both models.

i students

Holding students constant (Student = 1). The regression coefficient of income is 5.9843. This means that for every 1 unit ($1000) increase in income, a student's credit balance increases by $5.9843.

ii non-students.

Holding students constant (Student = 0). The regression coefficient of income is 5.9843. This means that for every 1 unit ($1000) increase in income, a student's credit balance increases by $5.9843.

(e) (practice)

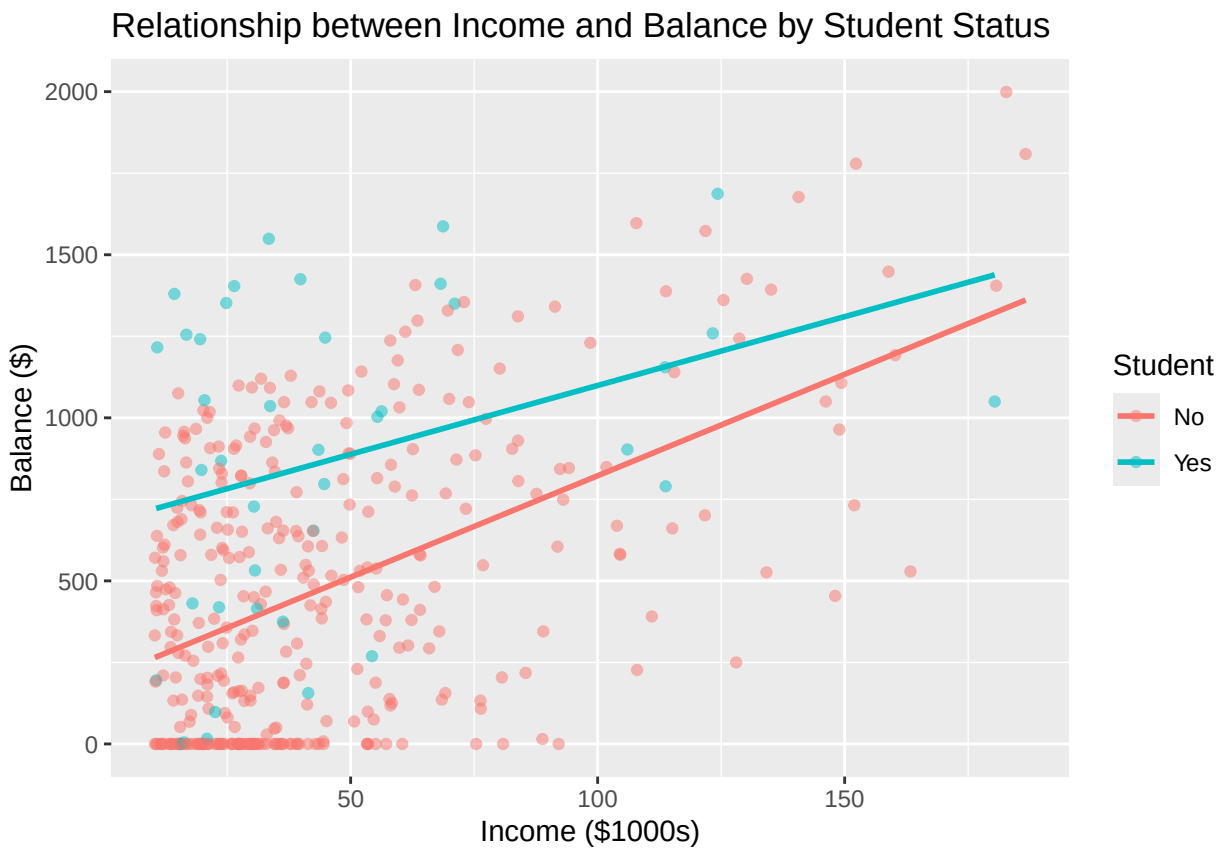
Notice that our model says that regardless of student status, the effect of Income on average Balance is the same. Do you think this is a reasonable constraint on our model?

Construct some plots to back up your answer.

This does not make sense because based on the plots below, it's clear that students carry a higher balance than non-students.

```r
library(ggplot2)

ggplot(Credit, aes(x = Income, y = Balance, color = Student)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship between Income and Balance by Student Status",
       x = "Income ($1000s)",
       y = "Balance ($)")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



(f) (practice)

One way we could relax this assumption is by incorporating interaction terms into our model. Specifically:

Yi = β_0 + β_1 Xi1 + β_2 Xi2 + β_3 Xi3 + ϵi

where X1= Income, X2 = Student, and X3 = Income × Student. Fit a model with an interaction term using the following code:

```
lm(Balance ~ Income + Student + Income:Student, data=Credit)
```

```
##
## Call:
## lm(formula = Balance ~ Income + Student + Income:Student, data = Credit)
##
## Coefficients:
##       (Intercept)             Income         StudentYes  Income:StudentYes
##           200.623              6.218            476.676             -1.999
```

Based on this model, write out the fitted model for students and write out the fitted model for non-students.

students = 200.623 + (6.218 - 1.999) * Income + 476.676

non_students = 200.623 + 6.218 * Income

(g) (practice)

Interpret the regression coefficient related to Income for the fitted models obtained in part (f).

For students, the income coefficient is $4.219. This means while holding all other predictors fixed, for every 1 unit ($1000) increase in income, a student's credit balance increases by $4.219

For non-students, the income coefficient is \$6.218. This means while holding all other predictors fixed, for every 1 unit (\$1000) increase in income, a student's credit balance increases by \$6.218

So, whether or not a person is a student does have an impact on their balance.